

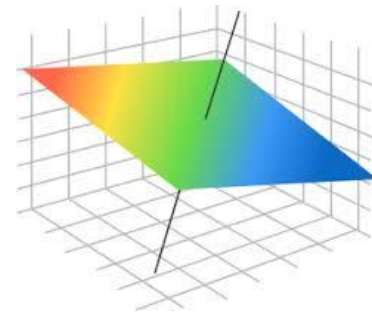
Dimension reduction

Lee-Ad Gottlieb

High dimension is bad!

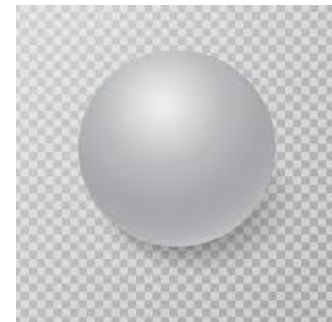
- For **learning**

- High dimensional objects (hyperplanes, balls) have high VC-dimension.
- Nets have VC-dimension exponential in d
- So sample will need to be large



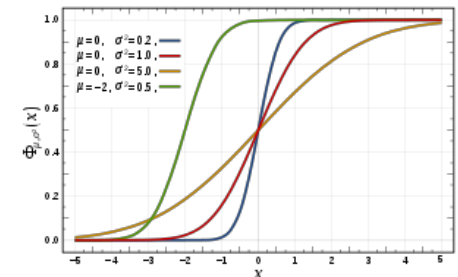
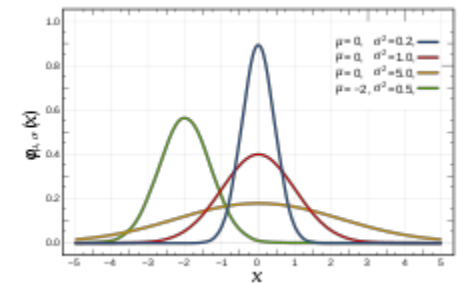
- For **computational tasks**

- Many tasks have runtimes that increase **exponentially** in the dimension
- Ex: exact nearest neighbor



Curse of dimensionality

- **Curse of dimensionality** (Bellman, 1961)
- Task: Estimating probability distribution functions
 - Probability density function
 - Cumulative distribution function
 - Probability mass function
- Problem:
 - Solution complexity
 - and required memory
 - grow **exponentially** with the dimension.



High dimension

- High dimension might also be unnecessary
 - Suppose we took 1000 measurements of a person's body.
 - Most of these measurements will be correlated
 - Can guess them without measuring
- Johnson & Lindenstrauss (1984)
 - Fundamental result in dimension reduction
 - Given n points in high dimension
 - Can reduce the dimension to $\log n$ without affecting distances much

JL Lemma

- Johnson-Lindenstrauss ('84):
 - Given a set of n Euclidean vectors $V \subset R^d$
 - there exists a **linear** function $f: V \rightarrow R^k$ for $k = \lceil \frac{8 \ln n}{\epsilon^2} \rceil$
 - Such that for all $x, y \in V$ and $0 < \epsilon \leq 1$
$$(1 - \epsilon)\|x - y\|_2 \leq \|f(x) - f(y)\|_2 \leq (1 + \epsilon)\|x - y\|_2$$

JL transform

- Example 1: The following random linear function realizes the JL-lemma:
- f is a multiplication matrix F of size $k \times d$
 - k is number of columns, d is number of rows
- F_{ij} is a Bernoulli random variable in
 - Takes values $\{-1, 1\}$ with probability $p = \frac{1}{2}$
 - (Later, we'll normalize F by $1/\sqrt{k}$)

$$\begin{array}{c} x \\ (8, 1, 6, 3) \end{array} \begin{array}{cc} F & f(x) \\ \begin{array}{|c|c|} \hline -1 & 1 \\ \hline 1 & -1 \\ \hline 1 & 1 \\ \hline -1 & 1 \\ \hline \end{array} & = (-4, 16) \end{array}$$

JL transform

- We'll prove that f preserves distances between points
- Take any points $x, y \in S$
 - Let $w = x - y$
 - Want to show that $\|f(x) - f(y)\|_2 \approx \|x - y\|_2$
 - Same as $\|f(w)\|_2 \approx \|w\|_2$
 - By linearity $f(x - y) = f(x) - f(y)$
 - Same as $\|f(w)\|_2^2 \approx \|w\|_2^2$
- So we need to show that the transform preserves norm of distance vectors w .

JL transform

- We need to show that the transform preserves norm of distance vectors w .
 - Let's start with a matrix F that's just a column vector

$$\begin{array}{ccc} w & F & f(w) \\ (8, 1, 6, 3) & \begin{array}{c} -1 \\ 1 \\ 1 \\ -1 \end{array} & = (-4) \end{array}$$

JL transform

- By construction, $f(w)^2 = (\sum_j F_j w_j)^2$
- For example, in 4 dimensions:
 - $f(w)^2 = (F_1 w_1 + F_2 w_2 + F_3 w_3 + F_4 w_4)^2$
 $= \sum_{i=1}^4 (F_i w_i)^2 + 2 \sum_{i < j} (F_i w_i \cdot F_j w_j)$
- In d dimensions:
 - $f(w)^2 = (F_1 w_1 + \dots + F_d w_d)^2$
 $= \sum_{i=1}^d (F_i w_i)^2 + 2 \sum_{i < j} (F_i w_i \cdot F_j w_j)$

JL transform

- By construction, $f(w)^2 = (\sum_j F_j w_j)^2$
- By linearity of expectation:

$$\begin{aligned} \bullet \quad E[f(w)^2] &= E[(\sum_{j=1}^d F_j w_j)^2] \\ &= E[\sum_{j=1}^d (F_j w_j)^2 + 2 \sum_{i < j} F_i w_i \cdot F_j w_j] \\ &= \sum_{j=1}^d E[(F_j w_j)^2] + 2 \sum_{i < j} E[F_i w_i \cdot F_j w_j] \\ &= \sum_{j=1}^d w_j^2 \\ &= \|w\|_2^2 \end{aligned}$$


$F_j^2=1$ $E[F_i F_j]=0$

JL transform

- Now suppose the matrix has k columns instead of one.
 - Let F_{ij} be the j -th row of the i -th column
 - By construction, $\|f(w)\|_2^2 = \sum_{i=1}^k (\sum_{j=1}^d F_{ij} w_j)^2$
- From the last slide, we have for any column i

$$E[(\sum_{j=1}^d F_{ij} w_j)^2] = \|w\|_2^2$$

w F $f(w)$

	-1	1	
	1	-1	
	1	1	
	-1	1	

(8, 1, 6, 3) = (-4, 16)

JL transform

- So we have

- $$\begin{aligned} E[\|f(w)\|_2^2] &= E[\sum_{i=1}^k (\sum_{j=1}^d F_{ij} w_j)^2] \\ &= \sum_{i=1}^k E[(\sum_{j=1}^d F_{ij} w_j)^2] \\ &= \sum_{i=1}^k \|w\|_2^2 \\ &= k \|w\|_2^2 \end{aligned}$$

- If we normalize matrix F by $\frac{1}{\sqrt{k}}$, then

$$E[\|f(w)\|_2^2] = \|w\|_2^2$$

JL transform

- Story until now:
 - For every $x, y \in S$
 - Each coordinate has $E[(f(x - y)_i)^2] = \|x - y\|_2^2$ (in *expectation*)
- What we really want:
 - For every $x, y \in S$
 - $\sum_i (f(x - y)_i)^2$ preserves $\|w\|_2^2$ up to a factor of $(1 \pm \epsilon)$ with *high probability*
 - If the probability is high enough $(1 - \frac{1}{2n^2})$, then it will preserve all n^2 distances in V at once:
 - $\Pr[\text{fails for one given pair}] \leq \frac{1}{2n^2}$
 - $\Pr[\text{fails for any pair}] \leq \frac{n^2}{2n^2} = \frac{1}{2}$ union bound
- We can show this with a Chernoff-like result:

JL transform

- For any vector w , we have that

$$\Pr[\|f(w)\|_2^2 > (1 + \epsilon)\|w\|_2^2] =$$

$$\Pr[\|f(w)\|_2^2 > (1 + \epsilon)E\|f(w)\|_2^2] < e^{-\frac{k}{2}(\frac{\epsilon^2}{2} - \frac{\epsilon^3}{3})}$$

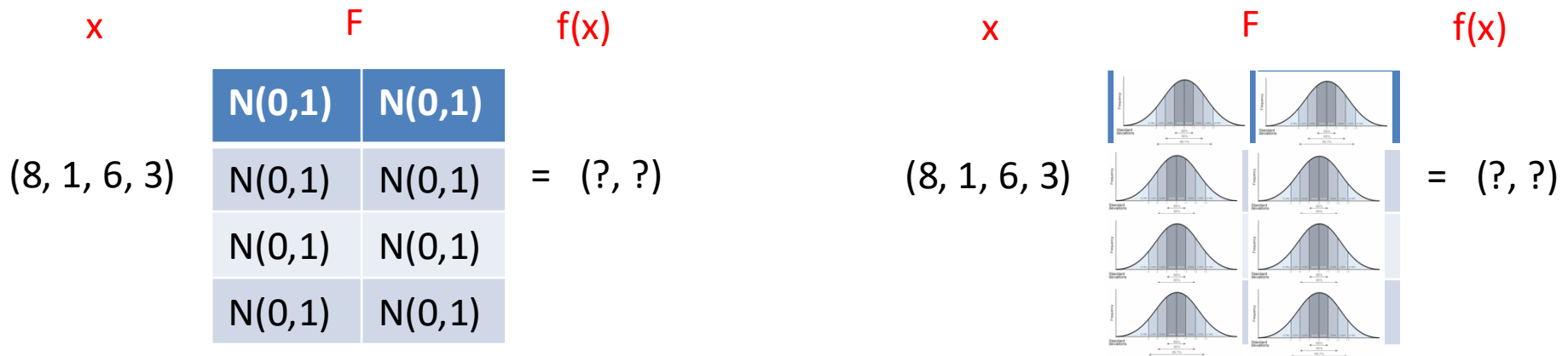
$$\Pr[\|f(w)\|_2^2 < (1 - \epsilon)\|w\|_2^2] =$$

$$\Pr[\|f(w)\|_2^2 < (1 - \epsilon)E\|f(w)\|_2^2] < e^{-\frac{k}{2}(\frac{\epsilon^2}{2} - \frac{\epsilon^3}{3})}$$

- So we take $k \sim \frac{\ln n}{\epsilon^2}$

JL transform

- Example 2: The following random linear function realizes the JL-lemma:
- f is a multiplication matrix F of size $k \times d$
- F_{ij} is a $(0,1)$ -Normal random variable
 - (Later, we'll normalize F by $1/\sqrt{k}$)



JL transform

- Let's assume F has only a single column
 - So $f(w) = \sum_i F_i w_i$
 - And $f(w)^2 = (\sum_i F_i w_i)^2$
- Normal distribution has the following properties:
 - $a \cdot N(0,1) = N(0, a^2)$
 - $N(0, c) + N(0, d) = N(0, c + d)$
 - So: $a \cdot N(0,1) + b \cdot N(0,1) = N(0, a^2 + b^2) = \sqrt{a^2 + b^2} \cdot N(0,1)$
- It follows that
 - $f(x) = \sum_i F_i w_i \sim \sum_i (w_i \cdot N(0,1)) = \|w\|_2 \cdot N(0,1)$
 - $f^2(x) = \|w\|_2^2 \cdot N^2(0,1)$

JL transform

- We proved that $f(x)^2 \sim \|w\|^2 \cdot N^2(0,1)$
 - What is $N^2(0,1)$?
 - Chi-square with 1 degree of freedom
- If matrix F has k columns
 - $\|f(w)\|_2^2 \sim \sum_{i=1}^k (\|w\|^2 \cdot N^2(0,1))$
$$= \|w\|^2 \cdot \sum_{i=1}^k N^2(0,1)$$
 - What is $\sum_{i=1}^k N^2(0,1)$?
 - Chi-square with k degrees of freedom

JL transform

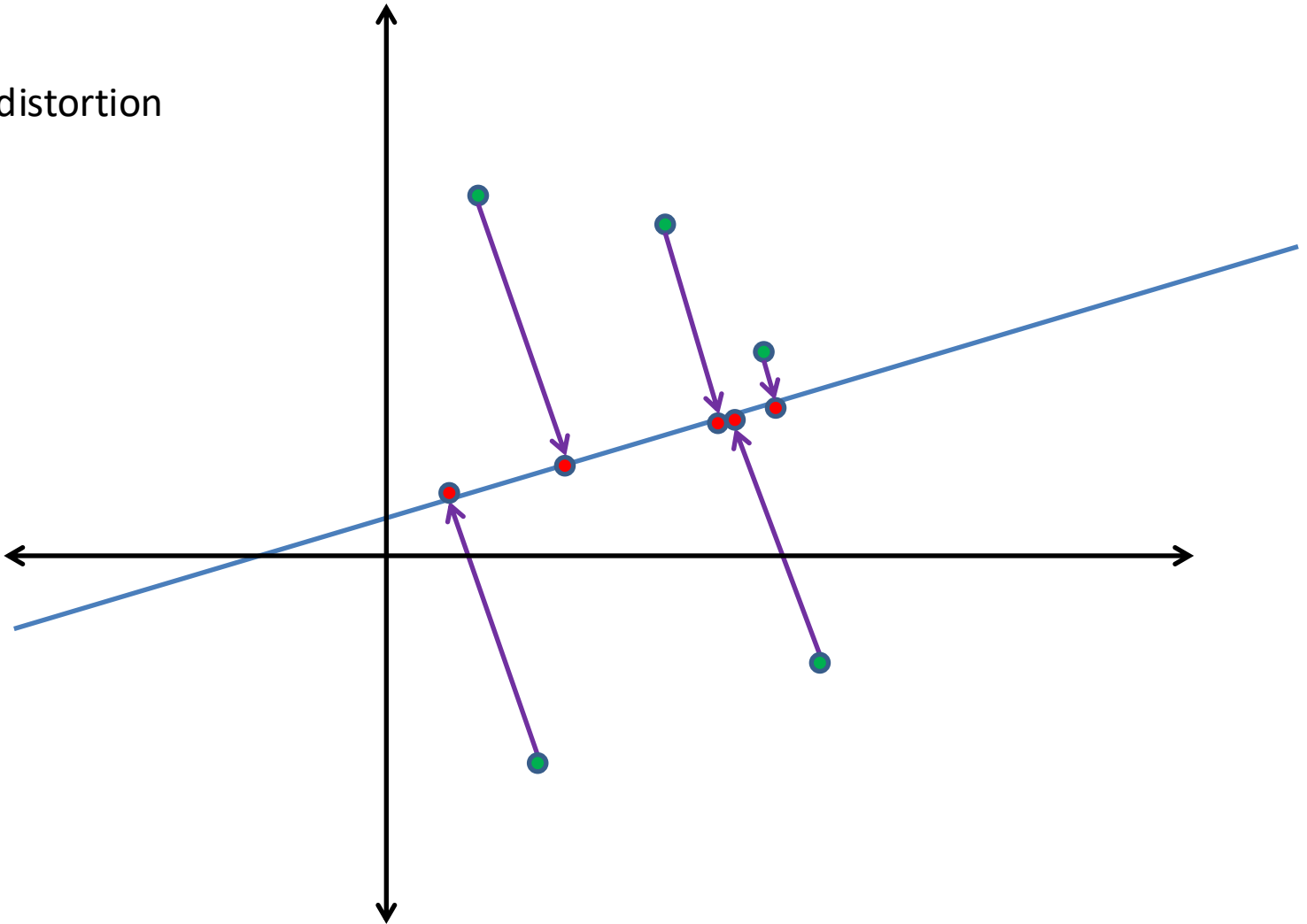
- Theorem: Let random variable x be sampled from a chi-squared distribution with k degrees of freedom and mean $E[x]$. Then,

$$\Pr[x > (1 + \epsilon)E[x]] < e^{-\frac{k}{2}(\frac{\epsilon^2}{2} - \frac{\epsilon^3}{3})}$$

- Take $k \geq \frac{4 \ln n}{\frac{\epsilon^2}{2} - \frac{\epsilon^3}{3}}$, and probability that one pair x, y is distorted is less than $\frac{1}{n^2}$
 - Holds for all point pairs
- done!

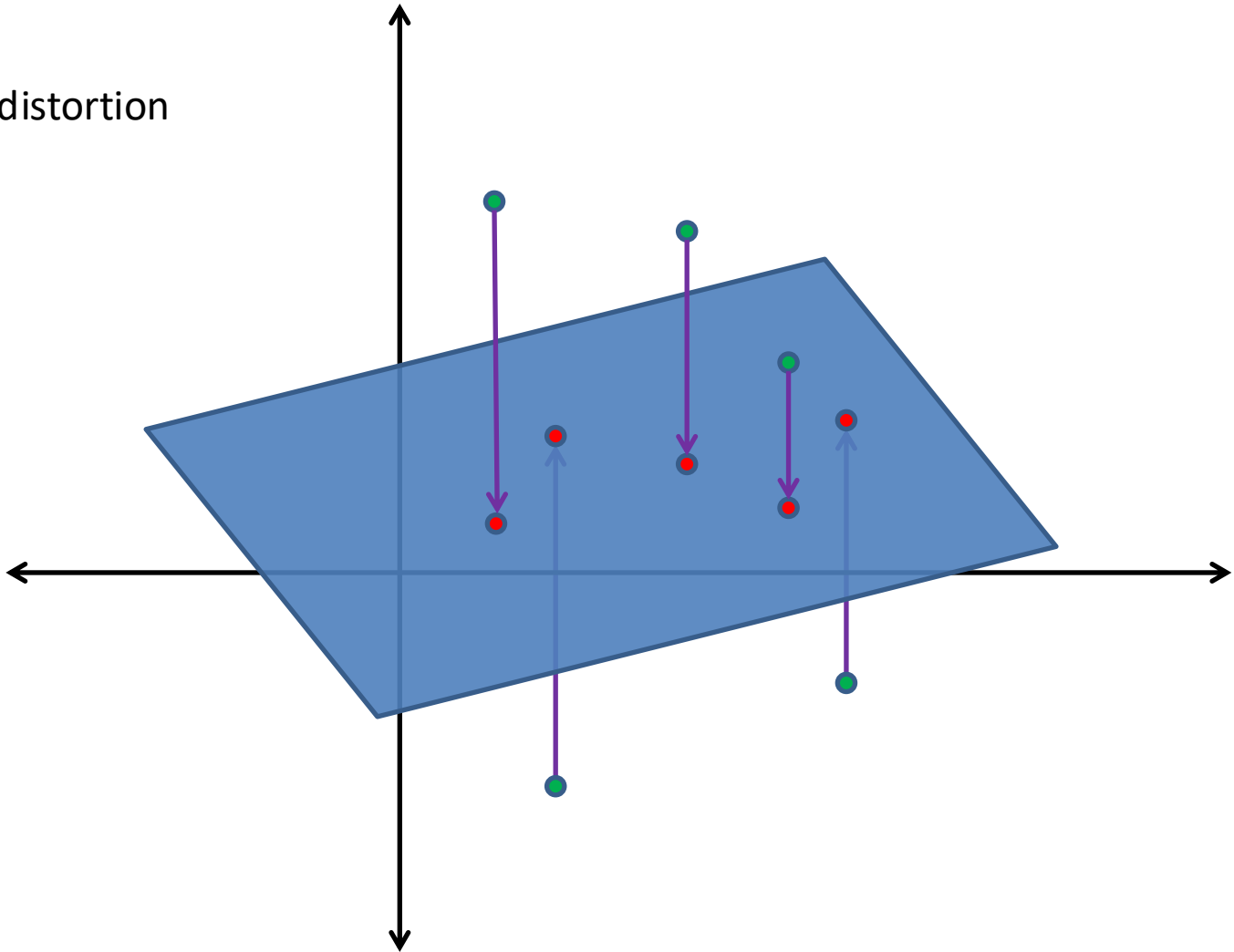
Johnson-Lindenstrauss into 1D

Measure:
Worst-case distortion



Johnson-Lindenstrauss into 2D

Measure:
Worst-case distortion

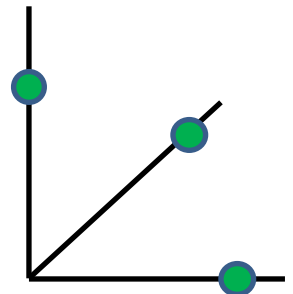


Lower-bound for JL

- How many dimensions are necessary to achieve distortion $(1 + \epsilon)$?
 - Can we do better than $k = \frac{8 \ln n}{\epsilon^2}$?

Lower-bound for JL

- Easy to show that l_2 requires $\Omega(\log n)$ dimensions
 - Volume argument on the n basis vectors
 - $\left(\frac{1}{\sqrt{2}}, 0, 0\right), \left(0, \frac{1}{\sqrt{2}}, 0\right), \left(0, 0, \frac{1}{\sqrt{2}}\right)$
 - Distance between every vector pair is 1
 - Reduce dimension, contraction at most $1/2$
 - So every point has distance $1/2$ to every other point $\rightarrow$ is the center of non-intersection balls of radius $1/4$.
 - $\frac{V(k, 1)}{V(k, \frac{1}{4})} = 4^k$
 - So must take $k \geq \log_4 n$



Lower-bound for JL

- How many dimensions are necessary to achieve distortion $(1 + \epsilon)$?
 - Dependence on $\log n$ unavoidable
 - What about dependence on ϵ^2 ?

- Noga Alon showed a lower-bound of:

$$k = \Omega\left(\frac{\log n}{\epsilon^2 \log \frac{1}{\epsilon}}\right)$$

- This leaves a gap of $\log(\frac{1}{\epsilon})$ between upper-bound and lower-bound
 - Resolving this gap was a long-standing open problem...

Lower-bound for JL

- In 2017, Jelani Nelson and Kasper Green-Larson showed a lower-bound of:

$$k = \Omega\left(\frac{\log n}{\epsilon^2}\right)$$

- This resolved the gap, and JL is tight up to constants

Lower-bounds for other l_p

- How many dimensions are necessary to achieve distortion $(1 + \epsilon)$, or even 2?
- l_1 requires $\Theta(n)$ dimensions
 - Andoni, Charikar, Neimann, Nguyen
- l_∞ requires $\Theta(n)$ dimensions
 - Proof: ...
- Open problem for other values of p .
 - Known upper-bound $O(n^2)$